

swtpcsim — FLEX on the SWTPC 6800

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The built-in `swtpc` machine is a Motorola 6800 with SWTBUG on its MP-S console and a DC-4 floppy controller (`dc4`) — the hardware TSC's **FLEX** operating system ran on. Each machine file here is that same machine with a FLEX boot disk mounted on drive 0, so it comes up at SWTBUG's `$` prompt ready to boot:

```
cd examples/swtpc/flex
swtpcsim flex2-40.toml
$                (SWTBUG prompt)
D                (SWTBUG disk boot)
```

`D` is SWTBUG's disk-boot command: it loads FLEX's cold-loader from track 0 and jumps into it. FLEX signs on and drops to its `+++` prompt. From there `CAT` lists the disk, and the usual FLEX commands work.

The six variants

Two FLEX releases, each in the three geometries the DC-4 supports. The board probes the geometry from the image **size**, so nothing in the `.toml` configures tracks or sides:

Machine file	Disk	FLEX	Geometry
<code>flex2-35.toml</code>	<code>FLEX2-35.DSK</code>	2.0	35-track single-sided, 88K
<code>flex2-40.toml</code>	<code>FLEX2-40.DSK</code>	2.0	40-track single-sided, 100K
<code>flex2-DS.toml</code>	<code>FLEX2-DS.DSK</code>	2.0	40-track double-sided, 200K
<code>flex3-35.toml</code>	<code>FLEX3-35.DSK</code>	3.0	35-track single-sided, 88K
<code>flex3-40.toml</code>	<code>FLEX3-40.DSK</code>	3.0	40-track single-sided, 100K
<code>flex3-DS.toml</code>	<code>FLEX3-DS.DSK</code>	3.0	40-track double-sided, 200K

Only one disk ships; the rest are one download away

`FLEX2-40.DSK` is the single image kept in the repository — it is what the `swtpc` quick start and the FLEX acceptance test boot. The other five machine files name sibling images that are **not** in the tree. They come from Mike Douglas's *FLEX Disk Images for the SWTPC 6800 Computer* on deramp.com — a folder of 6800-compatible FLEX 2.0/3.0 boot disks, with its own `-ReadMe.pdf` (kept here, also untracked):

<https://deramp.com/downloads/swtpc/software/FLEX/>

Download the ones you want and drop each `.DSK` beside its `.toml` — the names above are the names in that folder. `docs/sources.md` records the provenance and the exact SHA-256 of the shipped `FLEX2-40.DSK`.

Read-only by default

Every drive here is mounted `wriTEprotect = true`, so looking around FLEX can never dirty the image — a fresh clone stays a fresh clone, and there is no undo to need. To save files back (FLEX's `SAVE`, `NEWDISK`, `PCGET/PCPUT`, a file copy), drop the `wriTEprotect` line from the drive in the `.toml`, or copy the `.DSK` somewhere writable first.

Where the machine comes from

These files are the shipped `machines/swtpc.toml` with one disk mounted — see that file for why each board and address is what it is. For the sibling CP/68 boot example, see `examples/swtpc/cp68/`.